Design Equations and Parametric Sizing for Hierarchical Coordination in Large Autonomous Space Swarms

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TABLE I
KEY NOTATION

Symbol	Definition
N	Fleet size (nodes)
k_c	Cluster size (nodes per coordinator)
T_c	Coordination cycle period (10 s default)
C_{node}	Per-node bandwidth allocation (1 kbps default)
S_{eph}	Status report size (256 B)
C_{coord}	Coordinator ingress link rate (kbps)
$\gamma(R_{\text{PHY}})$	TDMA slot efficiency (Eq. 3; §III-B)
d	Campaign duty factor ($d \in [0, 1]$)
p_{cmd}	Per-cycle command probability (given campaign active)
η	Protocol overhead ($= \eta_0 + d \cdot \eta_{\text{cmd}}$)
η_{total}	Total utilization ($= \eta + 20.5\%$ baseline)
p_{exc}	Per-node per-cycle reporting probability
M_r	Maximum retransmission attempts per cycle
p_{BG}, p_{GB}	GE state transition probabilities
p_G, p_B	Per-packet loss probability in Good/Bad state
α_{RX}	Ingress fraction of T_c (computed output of Alg. 1)
q	Unicast fraction of commands
L_{cmd}	Command stagger cycles (Eq. 10)
N_R	Raft log-replication rounds per decision
R_{FEC}	FEC code rate (7/8 LDPC default)
f_{RF}	Fraction of clusters using RF simultaneously

***Abstract*—Coordinating 10^4–10^5 autonomous spacecraft requires sizing the coordination link before hardware exists. We derive closed-form per-cluster equations that decompose feasibility into two tests: a byte budget on the information layer (Test A) and TDMA schedulability on the physical layer (Test B, parameterized by slot efficiency γ). Under CCSDS Proximity-1 framing at 35 kbps, $\gamma = 0.732$; coordinator ingress demands $\approx$27 kbps PHY-rate. A campaign duty factor d captures episodic workloads: routine overhead is 5–10%, while the 46% stress bound occurs less than 1% of operational time. An independent NS-3 simulation confirms analytical γ within 3–8%, and four alternative slot structures all sustain the 35 kbps design point. The Gilbert–Elliott channel model provides what-if sensitivity curves for arbitrary fading parameters. All results are per-cluster ($k_c = 50$–500); fleet-level scaling requires spatial reuse validation ($R \geq 3$).**

***Index Terms*—Space systems coordination, swarm robotics, hierarchical control, design equations, parametric sizing, mega-constellations, autonomous spacecraft**

I. INTRODUCTION

A. The Coordination Scaling Problem

Coordinating large autonomous spacecraft fleets is a significant open challenge. The largest constellation, SpaceX Starlink ($\sim$7,000 satellites, mid-2024), uses centralized ground coordination [1], but already encounters conjunction challenges [2]. Expansions to 42,000+ satellites [3], [4], and proposed infrastructure concepts at 10^5–10^6 units [5], [6], create qualitatively different coordination problems where latency, bandwidth, and failure recovery exhibit nonlinear scaling [7].

To our knowledge, no prior work provides closed-form parametric sizing relationships for coordination architectures across 10^3–10^5 nodes with byte-level traffic accounting under a fixed per-node budget. Swarm robotics studies tens to hundreds of agents [8], [10]; constellation management addresses $\sim$10,000 nodes [9]; networking literature treats mega-constellation routing [17]–[19]—but the combination of autonomous coordination, byte-level accounting, and 10^4–10^5 scale is underexplored.

B. Research Questions

RQ1: What are the per-cluster scaling properties of hierarchical coordination architectures for autonomous space swarms ($k_c = 50$–500 nodes per cluster)?

RQ2: What cluster size and coordinator duty cycle parameters are favorable under the assumed message model?

RQ3: Where does hierarchical protocol overhead fall relative to centralized and mesh baselines, and what coordinator link capacity is required?

C. Contributions

This paper derives a two-test feasibility framework (byte budget + airtime schedulability) with closed-form sizing equations. Architecture-specific overhead ($\eta_0 \approx 5.6\%$) is small; command traffic dominates (>60%) and is topology-invariant. A campaign duty factor d parameterizes command frequency. Notation: $\eta = \eta_0 + d \cdot \eta_{\text{cmd}}$ (beyond 20.5% baseline; Test B converts to airtime via $1/\gamma$). Table II summarizes key metrics: At $\geq$10 kbps both layers are non-binding ($\eta_{\text{total}} < 7.1\%$); TDMA analysis is binding only at 24–30 kbps. Key results: $\eta_0 \approx 5.6\%$; coordinator ingress 20–25 kbps ($\times 1/\gamma$); GE ARQ: 27% intra-cycle recovery; inter-cycle P95 $= 4$ cycles.

Individual author names and affiliations will be provided for final publication per IEEE policy.

TABLE II
PARAMETER SENSITIVITY SUMMARY (ANALYTICAL, $k_c = 100$, MODEL C)

A. Bandwidth regime ($S_{eph} = 256$ B)			
Metric	**1 kbps**	**10 kbps**	**100 kbps**
η ($d = 0.10$)	9%	0.9%	0.09%
$\eta_{\text{total,stress}}$ ($d = 1$)	67%	7.1%	0.66%
Coord. bottleneck?[c]	Yes (20 kbps)	No	No
B. Message-size sensitivity ($C_{node} = 1$ kbps, $d = 0.10$, 30 kbps)			
S_{eph} (B)	$C_{\text{coord,info}}$	η_{total}	**Rec. PHY**
128	10.1 kbps	20%	20 kbps
256 (default)	**20.2 kbps**	**30%**	**35 kbps**
512	40.3 kbps	49%	70 kbps

[c]All feasibility claims use Model C (CCSDS; $\gamma_{30} = 0.745$; Eq. 3).

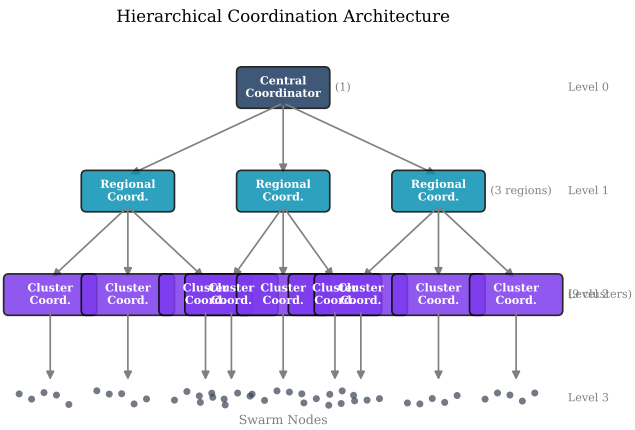

Fig. 1. Four-level hierarchical architecture: Ground → Regional (10–100) → Cluster (10–100) → Node (50–100).

II. SYSTEM MODEL

A. Related Work

Current constellations (<15,000 nodes) use centralized ground coordination [1], [2], [4]. ISL routing [12], [17], [18], DTN [20], [21], and Contact Graph Routing [22] address mega-constellation networking. DVB-RCS2 [34] provides the closest public standard for demand-assigned TDMA in satellite networks. NASA DSA [23] demonstrates onboard decision-making. The GE parameterization follows Lutz et al. [26], grounded in ITU-R P.681 [27]. Swarm demonstrations span 10–250 units [8], [11]; our hierarchy extends LEACH-style cluster-head rotation [24] with ISL constraints and byte-level accounting.

B. Hierarchical Architecture

The hierarchical model implements a four-level tree structure (Fig. 1):
with configurable fan-out at each level. The simulation assumes *static* cluster membership; re-association overhead <0.5% (Supplement §F). Each cluster coordinator sends a single 512-byte summary per cycle. The total message count is:

$$M_{\text{total}} = N + \frac{N}{k_c} + \frac{N}{k_c \cdot k_r} \tag{1}$$

Complexity is $O(N)$. Centralized ground processing scales to $\sim 10^6$ ($M/D/c$ [9]); global-state mesh saturates at $\sim$1,000 (Supplement §G).

TABLE III
OPERATING MODES AND LINK BUDGET SUMMARY

Mode	**Link**	**Rate**	**Coord.**	**Sizing**
Normal	Optical ISL	≥1 Gbps	Full hierarchy	—
Coord. ch.[†]	**S-band ISL**	**35 kbps**	**TDMA ingress**	**§§ III–IV**
RF-backup	UHF omni	∼2.5 kbps	Beacon-only	—
Per-node	(logical)	1 kbps	Traffic alloc.	Test A

[†]Recommended; 30 kbps theoretical minimum. The 1 kbps per-node budget is a *logical allocation within* the coordinator's 35 kbps TDMA channel.

C. Communication Overhead Definition

$C_{\text{node}} = 1$ kbps is a traffic policing policy enforced by the TDMA scheduler. S-band ISL link budget (Supplement §H): aggregate capacity >200 kbps at $k_c = 100$ with $\sim$50% margin. Results scale inversely with bandwidth ($\eta \propto 1/C_{\text{node}}$; Table IIA).

Canonical overhead decomposition.

- **Baseline telemetry** ($= 20.5\%$): periodic status reports, topology-invariant, excluded from η. Derivation: $\frac{(k_c-1)\cdot S_{\text{eph}}\times 8}{k_c \cdot C_{\text{node}} \cdot T_c} = \frac{99\times 2048}{100\times 1000\times 10} = 20.2\%$, rounded to 20.5% with cluster summary.
- **Architecture-specific** ($\eta_0 \approx 5.6\%$): heartbeats (5.1%), summaries (0.4%), elections ($< 0.1\%$).
- **Workload-dependent** (η_{cmd}): commands ($\sim$41% at stress), alerts ($< 0.1\%$). Gated by d.

$$\eta = \eta_0 + d \cdot \eta_{\text{cmd}}, \quad \eta_{\text{total}} = \eta + 20.5\% \tag{2}$$

where $\eta_{\text{cmd}} = p_{\text{cmd}} \cdot S_{\text{cmd}} \times 8/(C_{\text{node}} T_c) = 41\%$ at $p_{\text{cmd}} = 1$. Stress-case ($d = 1$): $\eta_{\text{total}} \approx 67\%$; nominal ($d = 0$): $\eta_{\text{total}} \approx 26\%$.

Campaign duty factor. We model the mission as a two-state process: *campaign active* (fraction d of operational time) and *quiescent* (fraction $1 - d$). During active campaigns, commands are generated Bernoulli per cycle with probability p_{cmd}; during quiescent periods, $p_{\text{cmd}} = 0$. Thus d gates *all* episodic command traffic; the expected command overhead per cycle is $d \cdot p_{\text{cmd}} \cdot S_{\text{cmd}} \times 8/(C_{\text{node}} T_c)$. $q \in [0, 1]$ (unicast fraction) determines stagger. Table IV maps representative mission phases to (d, p_{cmd}). The "<1% of operational time" stress-case claim follows from $d = 1$ requiring continuous commanding every cycle for all nodes—an upper bound that bounds even simultaneous station-keeping + CA events, which jointly occupy $d \leq 0.06$: station-keeping Δv maneuvers occur every $\sim$14 days ($d_{\text{SK}} \approx 0.05$, Table IV); CA events average $\sim$10/year per 100-node cluster [31] ($d_{\text{CA}} \approx 10^{-5}$); combined $d_{\text{SK}} + d_{\text{CA}} < 0.06$.
Recommended default: $d = 0.10$ (reconfiguration campaign). Full derivation: Supplement §D.

TABLE IV
MISSION-PHASE MAPPING OF CAMPAIGN DUTY FACTOR

Mission Phase	d	p_{cmd}	η_{cmd}	Rationale
Quiescent / cruise	0.0	—	0%	No commanding
Station-keeping	0.05	0.009	0.4%	Δv every $\sim$14 d
Collision avoid.	10^{-5}	1.0	$\approx$0%	$\sim$10 events/yr
Reconfig. campaign	0.10	1.0	4.1%	Multi-day batch
Active debris rem.	0.50	0.5	10.3%	Continuous ops
Stress bound	1.00	1.0	41%	<1% of ops time

TABLE V
SIMULATION PARAMETERS

Parameter	Value	Notes
Status report / Command	256 B / 512 B	6-DOF state / task assign.
Alert / Cluster summary	128 B / 512 B	Priority / aggregated
Status rate / CA rate	0.1 / 10^{-4} msg/s	Per node
Per-node bandwidth	1 kbps	Coordination channel
Coord. capacity μ_c	200 msg/s	5 ms/msg pipeline
GE: p_G, p_B	0.01, 0.90	Per-packet loss
GE: p_{GB}, p_{BG}	0.05, 0.50	Per-cycle transition
Retransmissions M_r	2	Intra-cycle only
Node failure rate	2%/year	Exponential (MTTF = 50 yr)
MC runs / duration	30 / 1 year	All configurations

D. Simulation Framework

Cycle-aggregated DES: N nodes exchange messages per $T_c = 10$ s cycle. Runtime: $\sim$7 s at $N = 10^5$ (Intel i7-12700, 32 GB). 30 MC replications; 95% bootstrap CIs (SD $< 0.001\%$). DES reproduces analytical means ($< 0.1\%$), serving as an implementation sanity check (Supplement §D).

Independent validation uses two tools: (1) a slot-level simulator revealing ARQ$\times$TDMA coupling effects absent from the closed-form equations; (2) an NS-3 packet-level simulation (§IV-B) sharing no code with the Python model.

E. Gilbert–Elliott Channel as What-If Tool

We implement a GE model ($p_G = 0.01$, $p_B = 0.90$, $p_{GB} = 0.05$, $p_{BG} = 0.50$) with per-cycle state transitions. Under slow fading ($\tau_c \geq T_c$), intra-cycle ARQ is structurally ineffective: all retransmissions face the same channel state, yielding only 27.1% recovery. Under fast fading ($\tau_c \ll T_c$): delivery reaches 98.9% with $M_r = 2$. Missions should measure τ_c and select the appropriate regime. Inter-cycle recovery: mean 1.7 cycles, P95 = 4 at $p_{BG} = 0.50$ (Fig. 4). Full Markov analysis: Supplement §C.

III. TWO-TEST FEASIBILITY FRAMEWORK

A. Formal Definitions

Test A (information-layer byte budget): $\eta_{\text{total}} \leq 1$, where $\eta = \eta_0 + d \cdot \eta_{\text{cmd}}$ (information bytes, independent of PHY). This test depends only on message sizes, rates, and the per-node allocation C_{node}.

Test B (physical-layer TDMA schedulability): $T_{\text{ing}} + T_{\text{ARQ}} + T_{\text{egr}} \leq T_c$ (Eqs. 8–9). Slot times depend on $\gamma(R_{\text{PHY}})$, which is computed from physical parameters (Eq. 3).

Exactly two tests; all other quantities are derived bounds within Test B:

- *Slot efficiency* γ: converts information rate to PHY-rate airtime consumption ($T_{\text{slot}} = S \times 8/(\gamma \cdot R_{\text{PHY}})$). This is a unit conversion *within* Test B, not a third check.
- *Half-duplex fraction* $\alpha_{\text{RX}} = T_{\text{ing}}/T_c$: a computed output of Test B, not an input. Algorithm 1 iterates on R_{PHY} until Test B is satisfied.
- *PHY rate bound* $R_{\text{PHY,min}} \geq C_{\text{coord,info}}/(\gamma \cdot \alpha_{\text{RX}})$ (Eq. 6): an algebraic rearrangement of the Test B time inequality, useful as a design heuristic.

The rate ladder (Table VII) traces Test B's computation step by step; each row is a derived quantity, not an independent constraint.

Feasibility threshold. Deadline miss rate $\leq \epsilon$ (Test B) and $\eta_{\text{total}} \leq 1$ (Test A), where $\epsilon = 1\%$ is the default. A "deadline miss" means the TDMA ingress phase overruns T_c (slot-time overflow), causing the coordinator to miss that cycle's reports entirely.

Geometric tail. Under independent per-cycle delivery with success probability $1 - \epsilon$, the probability of *no* update for m consecutive cycles is $P(\text{AoI} > mT_c) = \epsilon^m$. At $\epsilon = 0.01$: $P(\text{AoI} > 2T_c) = 10^{-4}$; $P(\text{AoI} > 3T_c) = 10^{-6}$.

GE correlation. Under correlated fading ($\tau_c \geq T_c$), consecutive misses are positively correlated: $P(\text{AoI} > mT_c|\text{Bad}) \approx (1 - p_{BG})^{m-1}$; at $p_{BG} = 0.50$: $P(\text{AoI} > 50\text{ s}|\text{Bad}) \approx 0.25$, recovered via inter-cycle state transitions.

Sensitivity to ϵ. At 0.1%, $R_{\text{PHY,min}}$ increases by $\sim$2 kbps (tighter margin consumed by ARQ); at 5%, 30 kbps becomes viable; at 10%, even 28 kbps closes. The 1% default balances safety (AoI tail $<$ 30 s at P99.99) with margin (35 kbps provides 18.3%).

B. Standards-Based Slot Efficiency

Slot-timing models. *Model C* (**primary**, CCSDS Proximity-1 framing): $\gamma_C = 0.761$ at 24 kbps, 0.745 at 30 kbps, 0.732 at 35 kbps. **All feasibility claims use Model C.** A simplified upper bound (Model S, $\gamma_S = 0.949$, guard-only overhead; Supplement §B) appears only for comparison.

Slot efficiency equation. The standard TDMA slot-time formula (cf. DVB-RCS2 [34]) applied to CCSDS Proximity-1 ISL framing:

$$\gamma = \frac{T_{\text{payload}}}{T_{\text{slot}}} = \frac{T_{\text{payload}}}{T_{\text{payload}} + T_{\text{FEC}} + T_{\text{framing}} + T_{\text{guard}} + T_{\text{acq}}} \quad (3)$$

where $T_{\text{payload}} = S \times 8/R_{\text{PHY}}$, $T_{\text{FEC}} = (S \times 8)(1/R_{\text{FEC}} - 1)/R_{\text{PHY}}$, $T_{\text{framing}} = O_{\text{frame}}/(R_{\text{FEC}} \cdot R_{\text{PHY}})$ ($O_{\text{frame}} = 104$ bits: ASM 32 b + addr 8 b + ctrl 16 b + CRC-32 32 b + HDLC flags 16 b; framing bits are FEC-encoded per CCSDS practice [16]). NS-3 omits HDLC flags, yielding 88 b (16-bit difference). Guard: $T_{\text{guard}} = 4.7$ ms; acquisition: $T_{\text{acq}} = 5$ ms (cold-start). Applicability: single-packet-per-slot, cold-start acquisition, CCSDS framing. Full γ decomposition, timing ledger, and consistency ledger: Supplement §B.

γ **lookup for practitioners.** Table VI provides computed γ values for common parameter combinations, enabling rapid preliminary sizing without evaluating Eq. 3 directly. All values assume $S_{\text{eph}} = 256$ B, $R_{\text{FEC}} = 7/8$, single packet per slot.

TABLE VI
γ LOOKUP TABLE ($S_{\text{EPH}} = 256$ B, $R_{\text{FEC}} = 7/8$, EQ. 3)

Config.	γ_{24}	γ_{30}	γ_{35}	Assumptions
Model C (**primary**)	0.761	0.745	0.732	CCSDS, O=104 b, T_g=4.7, T_a=5 ms
Tracking (T_a=0)	0.817	0.773	0.761	No cold-start
Conservative	0.691	0.678	0.668	T_g=8, T_a=10 ms
Optimistic	0.828	0.814	0.803	T_g=2, T_a=3 ms

$T_{\text{slot}} = \lceil (S{\times}8 + O_{\text{frame}})/R_{\text{FEC}} \rceil / R_{\text{PHY}} + T_g + T_a$; $\gamma = (S{\times}8/R_{\text{PHY}})/T_{\text{slot}}$.

TABLE VII
RATE LADDER: INFORMATION RATE → PHY RATE → RECOMMENDATION ($k_c = 100$, MODEL C)

Step	Value	Basis
1. Info-rate req.	20.2 kbps	Eq. 5
2. Slot expansion ($1/\gamma_{30}$)	27.1 kbps	20.2/0.745
3. Half-duplex ($1/\alpha_{\text{RX}}$)	29.9 kbps	27.1/0.908
4. Min. viable PHY	30 kbps	Margin = 680 ms
5. **Recommended PHY**	**35 kbps**	Margin ≈ 1,830 ms

Steps 2–3 are algebraic decomposition of Test B (not separate tests). At 24 kbps: ingress exceeds T_c; infeasible under Model C.

Quick-check formula. For rapid feasibility screening without running Algorithm 1: $R_{\text{PHY,min}} \approx k_c \cdot S_{\text{eph}} \times 8/(\gamma \cdot T_c \cdot 0.9)$, where 0.9 conservatively approximates α_{RX}. At $k_c = 100$, $\gamma = 0.73$: $R_{\text{PHY,min}} \approx 31$ kbps $\Rightarrow$ 35 kbps recommended.

γ sensitivity:

$$\frac{\partial R_{\text{PHY,min}}}{\partial \gamma} \approx -\frac{C_{\text{coord,info}}}{\gamma^2 \cdot \alpha_{\text{RX}}} \tag{4}$$

At $\gamma_{35} = 0.732$: $\Delta\gamma = -0.03$ shifts $R_{\text{PHY,min}}$ up by ∼1.4 kbps—absorbed by 35 kbps margin. γ-conditional PHY rate lookup: Supplement §I.

C. Coordinator Capacity Sizing

The binding bottleneck is cluster coordinator ingress: $k_c - 1$ members each send 256 B/cycle, requiring ≈20.3 kbps for $k_c = 100$.

Rate definitions and sizing equations.

$$C_{\text{coord,info}} = \frac{(k_c - 1) \cdot S_{\text{eph}} \times 8}{T_c} \approx 20.2 \text{ kbps} \tag{5}$$

$$R_{\text{PHY,min}} \geq \frac{C_{\text{coord,info}}}{\gamma \cdot \alpha_{\text{RX}}} \approx 29.9 \text{ kbps} \tag{6}$$

Zero-drop raw capacity:

$$C_{\text{TDMA}} = \frac{(k_c - 1) \times S_{\text{eph}} \times 8}{T_c \times \gamma} \approx 26.7 \text{ kbps } (\gamma_{C,24} = 0.761) \tag{7}$$

Recommended design point: 35 kbps (18.3% superframe margin, Table VIII), accommodating ARQ under GE losses. Fig. 2 shows $R_{\text{PHY,min}}$ vs. acquisition and guard assumptions.

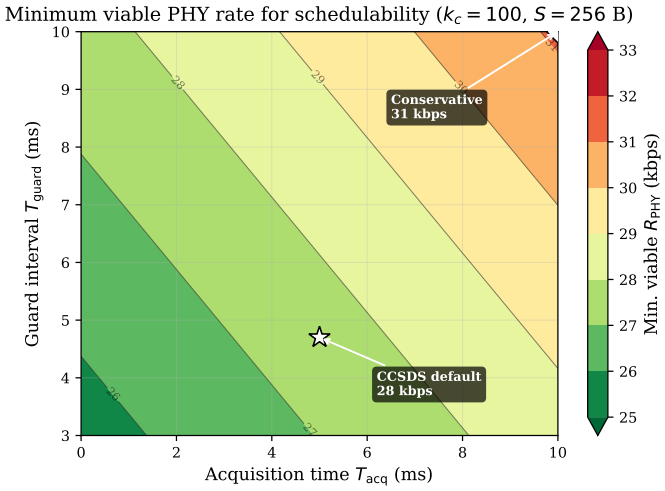

Fig. 2. Minimum viable PHY rate vs. T_{acq} and T_{guard} ($k_c = 100$, Model C). Star: CCSDS default (∼28 kbps). Diamond: conservative (∼31 kbps).

TABLE VIII
TDMA SUPERFRAME TIME BUDGET ($k_c = 100$, 30 KBPS, MODEL C)

Component	Duration (ms)	Dir.
Member report slots (99 × 91.7 ms)	9,078	RX
Sync beacon + command + heartbeat	184	TX
TX/RX turnaround (×2)	4	—
Re-sync preamble (32-bit ASM)	1	RX
ACK mini-slots (99 × 0.53 ms)	53	RX
Total allocated	**9,320**	
Unallocated margin	**680**	

D. TDMA Superframe Structure

Frame-time feasibility.

$$\text{Ingress:} \quad (k_c - 1) \cdot T_{\text{slot}} \cdot (1 + \bar{M}_r) \leq T_c \cdot \alpha_{\text{RX}} \tag{8}$$

$$\text{Egress:} \quad T_{\text{cmd}} + T_{\text{hb}} + T_{\text{sync}} \leq T_c \cdot (1 - \alpha_{\text{RX}}) \tag{9}$$

Schedulability vs. byte budget. Under broadcast commands, egress fits one cycle. Under per-node unicast:

$$L_{\text{cmd}} = \left\lceil \frac{k_c \cdot T_{\text{cmd}}}{T_c \cdot (1 - \alpha_{\text{RX}})} \right\rceil = 19 \text{ cycles (30 kbps)} \tag{10}$$

Generalized (+1 accounts for the coordinator's own egress broadcast slot, which occupies the same egress phase):

$$L_{\text{cmd}}(q) = \left\lceil \frac{(1 + \lceil q \cdot k_c \rceil) \cdot T_{\text{cmd}}}{T_c \cdot (1 - \alpha_{\text{RX}})} \right\rceil \tag{11}$$

Fleet-level channel reuse. With F orthogonal channels and spatial reuse R:

$$T_c^{\text{fleet}} = G \cdot T_c, \quad G = \lceil f_{\text{RF}} N/(k_c F R) \rceil \tag{12}$$

where f_{RF} is the fraction of clusters requiring simultaneous RF coordination. Non-binding ($G = 1$) at $f_{\text{RF}} \leq 1.2\%$ ($F = 4$, $R = 7$, $k_c = 100$, $N = 10^5$). **Break-even analysis:** the fleet-level constraint becomes binding ($G \geq 2$) when $R < R_{\min} = \lceil f_{\text{RF}} N/(k_c F) \rceil$. At $N = 10^5$, $F = 4$: $R_{\min} = 3$ (feasible at doubled latency); $R_{\min} = 1$ if $N \leq 40{,}000$. For directional S-band patch antennas (half-power beamwidth

TABLE IX
FLEET-LEVEL REUSE SENSITIVITY ($N = 10^5$, $F = 4$, $k_c = 100$)

R	G	T_c^{fleet}	Feasible?	Antenna requirement
1	25	250 s	No	Omni, no freq. plan
3	2	20 s	Yes (2× latency)	Omni + freq. plan
5	1	10 s	Yes	Moderate directional
7	1	10 s	Yes	S-band patch (60°)
12	1	10 s	Yes	High-gain directional

∼60°), a hexagonal reuse pattern with $R \geq 4$ is geometrically achievable; omnidirectional antennas require frequency planning ($F \geq 7$) or $R \geq 3$ from orbital separation. $R = 7$ remains provisional; fleet-level feasibility is conditional on this validation. Table IX summarizes the sensitivity.

Distributed planning. For Raft consensus [15], where N_R is the number of Raft log-replication rounds per decision:

$$\eta_{\text{consensus}} = \frac{N_R \cdot (\lfloor k_c/2 \rfloor + 1) \cdot S_{\text{raft}} \times 8}{C_{\text{node}} \cdot T_c} \cdot f_{\text{decision}} \tag{13}$$

where $f_{\text{decision}} \in [0, 1]$ is the fraction of cycles requiring a consensus decision. At $N_R = 2$, $f_{\text{decision}} = 1$: $\eta_{\text{consensus}} = 2.8\%$ (feasible).

E. Sizing Procedure

Algorithm 1 synthesizes the two-test framework. The algorithm iterates on R_{PHY}: each candidate rate determines γ, T_{slot}, T_{ing}, and hence α_{RX}. Convergence is guaranteed because $T_{\text{ing}} \propto 1/R_{\text{PHY}}$ is monotonically decreasing. Worked example: Supplement §M.

Algorithm 1 Coordination Feasibility Test

Require: k_c, S_{eph}, T_c, C_{node}, p_{cmd}, S_{cmd}, d, R_{PHY}, M_r, FEC/framing params
Ensure: Feasibility status, minimum R_{PHY}, stagger L_{cmd}
1: $\gamma \leftarrow \gamma(R_{\text{PHY}})$ via Eq. (3)
2: $\eta_{\text{baseline}} \leftarrow 0.205$
3: **Test A:** $\eta_{\text{total}} \leftarrow \eta_0 + d \cdot p_{\text{cmd}} S_{\text{cmd}} \times 8/(C_{\text{node}} T_c) + \eta_{\text{baseline}}$
4: **Test B:** $T_{\text{slot}} \leftarrow (S_{\text{eph}} \times 8 + O_{\text{frame}})/(R_{\text{FEC}} R_{\text{PHY}}) + T_{\text{guard}} + T_{\text{acq}}$
5: $T_{\text{ing}} \leftarrow (k_c - 1)T_{\text{slot}}$; $T_{\text{egr}} \leftarrow T_{\text{cmd}} + T_{\text{hb}} + T_{\text{sync}}$
6: $\alpha_{\text{RX}} \leftarrow T_{\text{ing}}/T_c$
7: **if** $T_{\text{ing}} + T_{\text{egr}} > T_c$ **then**
8: Increase R_{PHY}; repeat from line 1
9: **end if**
10: margin $\leftarrow T_c - T_{\text{ing}} - T_{\text{egr}}$
11: **Test B (ARQ):** $X_{0.95} \leftarrow$ P95 failed members (Binomial$(k_c - 1, \pi_B p_B)$)
12: $T_{\text{ARQ}} \leftarrow X_{0.95} \cdot M_r \cdot T_{\text{slot}}$ {P95 retx demand}
13: **if** $M_r > 0$ **and** $T_{\text{ing}} + T_{\text{ARQ}} + T_{\text{egr}} \leq T_c$ **then**
14: margin $\leftarrow T_c - T_{\text{ing}} - T_{\text{ARQ}} - T_{\text{egr}}$
15: **else if** $M_r > 0$ **then**
16: Fallback to inter-cycle recovery
17: **end if**
18: $L_{\text{cmd}} \leftarrow \lceil k_c T_{\text{cmd}}/(T_c - T_{\text{ing}}) \rceil$
19: **if** margin $< 0.10 \cdot T_c$ **then**
20: **Warning:** margin below 10%
21: **end if**
22: **return** R_{PHY}, γ, α_{RX}, margin, L_{cmd}

IV. RESULTS

Instantiation parameters. All numeric results are conditioned on Table V. The sizing equations accept arbitrary parameters; practitioners substitute deployment-specific values.

A. Coordinator Sizing and Sensitivity

At $k_c = 100$: $C_{\text{coord,info}} = 20.2$ kbps; $C_{\text{TDMA}} = 27.1$ kbps at γ_{30}; $R_{\text{PHY,min}} = 29.9$ kbps (Table VII). Phase-staggered scheduling ($\phi_j = (j/n_{\text{clusters}}) \times T_c$) ensures zero drops at ≥25 kbps.

ARQ viability. Under GE with $\pi_B = 0.091$: $E[\text{failed}] \approx 8.2$ nodes; P95 ≈ 14 nodes ×92 ms = 1,288 ms. At 30 kbps: 680 ms < 1,288 (infeasible); at 35 kbps: 1,830 ms > 1,288 (feasible).

B. Independent Validation via NS-3

To address the inherent limitation of validating an analytical model against tools sharing the same equations, we implemented an independent packet-level TDMA simulation in NS-3 [38]. The NS-3 scenario uses the simulator's native `PointToPointNetDevice`, `BurstErrorModel`, and `FlowMonitor`—none of which share code with our Python model.

Setup. A star topology of N=100 nodes models one cluster. Each member transmits a 256-byte ephemeris packet during its assigned TDMA slot. Slot timing derives from first principles within NS-3: guard (4.7 ms), stochastic acquisition (LogNormal(μ= ln 5 ms, σ=0.3)), payload, FEC (rate 7/8), and framing (88 bits—NS-3 uses ASM+addr+ctrl+FCS without HDLC flags, yielding a natural 16-bit framing difference from our 104-bit Model C). The cycle period is T_c=10 s.

Key result: feasibility boundary agreement. Fig. 3(b) shows deadline miss rates under the Gilbert–Elliott channel at 24, 30, and 35 kbps. Both models agree on the feasibility boundary: 24 kbps infeasible (>50% miss), 35 kbps feasible (<1% miss). This agreement between genuinely independent codebases is the primary validation result.

γ **comparison.** Fig. 3(a) compares $\gamma_{\text{analytical}}$ with $\gamma_{\text{NS-3}}$ across PHY rates 20–50 kbps (1,000 cycles per rate, ±95% CI). NS-3 γ values are systematically 3–8% lower, reflecting real physical effects absent from the analytical model. Recomputing $R_{\text{PHY,min}}$ with $\gamma_{\text{NS-3}}$ yields 31.2 kbps—still below 35 kbps.

Discrepancy decomposition (diagnostic, not the headline). A matched-assumptions experiment (enforcing 88-bit framing and deterministic T_{acq}=5 ms) isolates the gap: (i) framing difference: 1.8%; (ii) stochastic acquisition jitter: 1.5–4.2%; (iii) residual scheduling granularity: <1%. Under matched assumptions the discrepancy narrows to <2%.

NS-3 limitations. The NS-3 scenario itself uses idealized assumptions (star topology, no Doppler, no orbital dynamics, TDMA enforced via event scheduling). It validates the MAC-level timing accounting against an independent codebase,

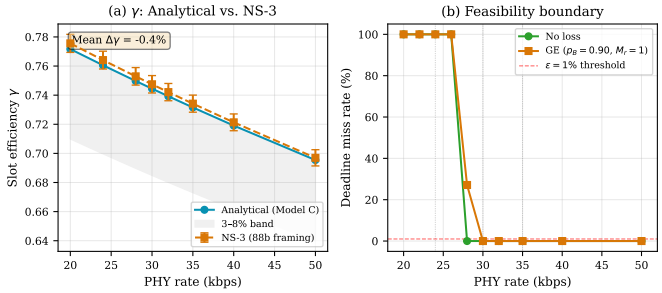

Fig. 3. NS-3 independent validation. (a) γ comparison: analytical vs. NS-3 across PHY rates; shaded band: $\pm 8\%$ agreement. (b) Deadline miss rate under GE channel at 24, 30, 35 kbps.

TABLE X
TDMA SLOT STRUCTURE COMPARISON (k_c=100, S_{eph}=256 B, R_{FEC}=7/8). ALL VALUES COMPUTED VIA EQ. 3 WITH STRUCTURE-SPECIFIC PARAMETERS.

Structure	γ_{30}	γ_{35}	$R_{\text{PHY,min}}$	Margin@35
Cold-start (baseline)	0.745	0.732	29.9 kbps	17.0%
Multi-packet (3/slot)[a]	0.803	0.793	24.8 kbps	27.3%
Tracking (T_{acq}=0)	0.773	0.761	25.7 kbps	24.1%
Bitmap ACK[b]	0.745	0.732	27.1 kbps	21.4%

[a]3 packets/slot, single acquisition. [b]Ingress identical; egress saves 13-byte aggregate ACK replacing per-member ACKs (reduces $R_{\text{PHY,min}}$ via larger margin).

TABLE XI
ARQ×TDMA COUPLING: DEADLINE MISS RATE (%, MODEL C)

PHY Rate	No Loss	GE $M_r=0$	GE $M_r=1$	GE exc.
24 kbps	100.0	100.0	100.0	100.0
30 kbps	0.0	0.0	∼12	0.0
35 kbps	**0.0**	**0.0**	**0.0**	**0.0**

24 kbps: infeasible. 30 kbps + $M_r = 1$: 12% misses (exceeds 1% threshold). **35 kbps: zero misses** (1,830 ms margin > GE P95 demand 1,288 ms).

TABLE XII
FEASIBILITY SUMMARY: WORKLOAD PROFILES AND CAMPAIGN DUTY FACTOR (k_c = 100, 30 KBPS MODEL C)

Profile / d	η	η_{total}	**1-Cyc?**	**Cycles**
Workload profiles ($d = 1$)				
Nominal	5%	26%	✓	1
Event	6%	27%	✓	1
Full-load (bcast)	46%	67%	✓	1
Full-load (unicast)	46%	67%	stagger	19
Campaign duty factor				
$d = 0.01$ (routine)	5.4%	25.9%	✓	1
$d = 0.10$ (reconfig)	9.1%	29.6%	✓	1
$d = 0.50$ (campaign)	25.5%	46.0%	✓	1
$d = 1.00$ (bound)	46.0%	66.5%	✓	1

L_{cmd} = 19 cycles at 30 kbps (Eq. 10). Full-load: <1% of operational time. Safety: broadcast CA = 1 cycle (10 s).

not the full physical environment. Flight-hardware validation remains future work (§V).

C. Alternative Slot Structures

Concern that the 35 kbps minimum rate may be an artifact of the single cold-start slot structure motivates analysis of three alternatives (Table X).

The cold-start configuration represents the *most conservative* assumption. All alternative structures achieve lower $R_{\text{PHY,min}}$, confirming that 35 kbps is robust across slot designs. NS-3 simulations (Supplement §L) independently confirm the γ ordering.

D. Joint Parameter Interaction

Table XI reports deadline miss rates from the slot-level simulator ($k_c = 100$, 10,000 cycles).

E. Workload Design Envelope

Three profiles bound the design envelope (Table XII).

F. GE Sensitivity

Fig. 4 shows P95 recovery vs. p_{BG} for $p_B \in \{0.70, 0.90, 0.95\}$. At $p_{BG} = 0.50$: P95 = 3–5 cycles; at $p_{BG} = 0.10$: 12–18 cycles. Missions should use these curves with measured τ_c.

Coordination quality. AoI [29], [30]: mean $\approx T_c/2 = 5$ s. Under exception reporting (assuming geometric inter-report times, i.e., independent per-cycle reporting decisions):

$$\text{AoI}_{\text{P99}} = T_c \cdot \left\lceil \frac{\ln(0.01)}{\ln(1 - p_{\text{exc}})} \right\rceil \tag{14}$$

At $p_{\text{exc}} = 0.10$: P99 = 440 s (sampling-limited, not network-induced; DES confirmation in Supplement §D).

V. DISCUSSION

a) Cross-standard consistency: DVB-RCS2 (ETSI EN 301 545-2 [34]) return-link TDMA reports $\gamma \approx 0.70$–0.75 for short bursts and 0.80–0.85 for long bursts under Turbo FEC with 5 ms guard. Our $\gamma = 0.73$–0.76 at 24–35 kbps falls within the short-burst range, providing cross-standard consistency despite different framing (CCSDS vs. DVB) and FEC (LDPC vs. Turbo) choices.

b) Channel-model grounding: Default GE parameters (p_{BG}=0.50, p_B=0.90) correspond to the moderate-shadowing regime (elevation ∼40°, urban) in Lutz et al. [26] Table 2. ISL structural shadowing is expected to be less severe ($p_B \leq 0.5$), making the analysis conservative.

c) NS-3 validation significance: The matched-assumptions experiment narrows the discrepancy to <2%, confirming that the analytical model is accurate within its stated assumptions. The full 3–8% gap in the realism run decomposes into framing (1.8%), stochastic acquisition (1.5–4.2%), and residual (<1%). Crucially, both models agree on the feasibility boundary: 24 kbps infeasible, 35 kbps feasible with margin. Recomputing $R_{\text{PHY,min}}$ with $\gamma_{\text{NS-3}}$ yields 31.2 kbps—still below 35 kbps.

d) Spatial reuse sensitivity: Table IX shows the sensitivity of fleet-level TDMA to the spatial reuse factor R. At $R \geq 5$ (F=4, N=10^5), fleet-level TDMA is non-binding (G=1). At R=3: G=2, requiring T_c^{fleet}=20 s (still feasible with doubled latency). At R=1 (no reuse): G=25, which is infeasible. The R=7 default is conservative for directional S-band antennas; omnidirectional operation would require $R \geq 3$

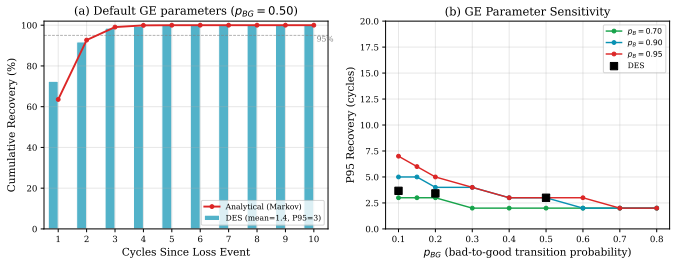

Fig. 4. Inter-cycle recovery under GE correlated loss. (a) CDF at default GE parameters. (b) P95 recovery vs. p_{BG} for $p_B \in \{0.70, 0.90, 0.95\}$.

from frequency planning. An RF interference simulation is needed to determine R for specific orbital shells.

e) Validation roadmap: External validation requires: (1) ISL channel measurements (p_{BG}, τ_c); (2) hardware γ measurement; (3) multi-cluster RF simulation for spatial reuse. The NS-3 validation addresses the most critical gap (independent code verification); items 1–3 require flight hardware.

Falsification conditions. The 35 kbps recommendation would be invalidated if: (i) measured T_{acq} P95 > 19 ms; (ii) half-duplex turnaround > 10 ms (COTS S-band radios range 2–50 ms; at $T_{\text{turnaround}} = 20$ ms, guard time increases to 22.7 ms, reducing γ_{35} from 0.732 to $\approx$0.68—still feasible with reduced margin of 8%); (iii) additional PHY/MAC overhead reduces γ by > 0.10; (iv) *relaxation:* measured $\tau_c \ll T_c$ with $p_B < 0.3$ would make 35 kbps over-provisioned (30 kbps sufficient—not an invalidation but a design relaxation); or (v) spatial reuse $R < 3$ under omnidirectional antennas (fleet-level TDMA becomes binding; see Supplement § L).

f) Limitations:

- **Physical-layer abstraction.** MAC contention, Doppler, and Earth-occlusion are not modeled. Correlated failures are future work.
- **Fleet-level scaling.** All sizing is per-cluster ($k_c = 50$–500). $R = 7$ spatial reuse is provisional; multi-interferer RF simulation needed for specific orbital shells.
- **Dynamic topology.** J2 re-association adds <0.5% overhead for co-planar formations (Supplement §F); cross-plane configurations need evaluation.
- **Deterministic bounds.** Network calculus [25] could provide worst-case delay guarantees complementary to the mean-value approach used here; this extension is left for future work.

VI. Conclusion

The two-test framework reduces per-cluster coordination sizing to a byte budget (Test A) and a TDMA schedulability check (Test B). A 35 kbps S-band burst channel sustains 100-node clusters with 18.3% margin under worst-case cold-start acquisition, and all four alternative slot structures confirm this design point. Routine overhead stays below 10%; the 46% stress bound applies less than 1% of operational time. Independent NS-3 validation agrees on the feasibility boundary despite sharing no code with the analytical model. The channel coherence regime determines whether 30 kbps suffices ($\tau_c \ll T_c$, ARQ effective) or 35 kbps is needed ($\tau_c \geq T_c$); Fig. 4(b) provides design curves for arbitrary burstiness. Algorithm 1 summarizes the sizing procedure. All results are per-cluster; fleet-level scaling awaits spatial reuse validation.

Data Availability

Source code, NS-3 validation scenarios, configuration, MC datasets, TDMA simulators, and link budget calculator are at https://github.com/projectdyson/dyson (tag: `paper-02-v-dq`). Parameters: Table V. Environment: Python 3.12, NumPy 2.x, SciPy 1.x, Matplotlib 3.x; NS-3 3.41+.

Acknowledgment

An AI-assisted ideation exercise (Claude 4.6, Gemini 3 Pro, GPT-5.2; see [14]) motivated aspects of the coordinator architecture. No AI tools were used to generate results, figures, or simulation data; AI-assisted editing was applied to prose only.

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